

Homogeneous Linear DE with constant co-efficients:

The equation of the form $a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1 \frac{dy}{dx} + a_0 y = 0$ is called homogeneous linear DE with constant co-efficients of order n .

Auxiliary Equation: If $\frac{dy}{dx} = m e^{mx}$, $\frac{d^2 y}{dx^2} = m^2 e^{mx}$, $\dots$, $\frac{d^n y}{dx^n} = m^n e^{mx}$ then the above eqn becomes

$$(a_n m^n + a_{n-1} m^{n-1} + \dots + a_1 m + a_0) e^{mx} = 0$$

where $e^{mx} \neq 0$

The equation $a_n m^n + a_{n-1} m^{n-1} + \dots + a_1 m + a_0 = 0$ is called the Auxiliary equation.

General solution for this type:

Case I: If all the roots of the A.E are distinct then the general solution will be $y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \dots + c_n e^{m_n x}$

Case II: If all the roots of the A.E are not distinct
i.e. $m_1 = m_2 = \dots = m_n$ then the general soln is

$$y = (c_1 + c_2 x + \dots + c_n x^{n-1}) e^{m_1 x} + c_{n+1} e^{m_2 x} + \dots + c_n e^{m_n x}$$

Case III: If all the roots of the A.E are non-real
then the general solution is

$$y = e^{\alpha x} [c_1 \cos \beta x + c_2 \sin \beta x] \quad [m = \alpha \pm i\beta]$$

~~Q~~ Solve $(D^2 + D - 6)y = 0$

Soln: let $y = e^{mx}$

$$\therefore (D^2 + D - 6)e^{mx} = 0$$

$$\Rightarrow (m^2 + m - 6)e^{mx} = 0$$

$$\therefore \text{A.E. } m^2 + m - 6 = 0$$

$$\Rightarrow m^2 + 3m - 2m - 6 = 0$$

$$\Rightarrow m(m+3) - 2(m+3) = 0$$

$$\Rightarrow (m+3)(m-2) = 0$$

$$\therefore m = 2, -3$$

The general soln is $y = c_1 e^{-3x} + c_2 e^{2x}$

~~Q~~ Solve $(D^2 - 4D + 4)y = 0$

Soln: let, $y = e^{mx}$

$$\therefore (D^2 - 4D + 4)e^{mx} = 0$$

$$(m^2 - 4m + 4)e^{mx} = 0$$

$$\therefore A.E \Rightarrow m^2 - 4m + 4 = 0$$

$$\Rightarrow m^2 - 2m - 2m + 4 = 0$$

$$\Rightarrow m(m-2) - 2(m-2) = 0$$

$$\Rightarrow (m-2)(m-2) = 0$$

$$\therefore m = 2, 2$$

The general solution is $y = (C_1 + C_2 x)e^{2x}$

~~Ex~~ solve $(D^2 - 4D + 13)y = 0$

Soln: let $y = e^{mx}$

$$\therefore (D^2 - 4D + 13)e^{mx} = 0$$

$$(m^2 - 4m + 13)e^{mx} = 0$$

$$\therefore A.E \text{ is } m^2 - 4m + 13 = 0$$

$$\therefore m = \frac{4 \pm \sqrt{16 - 52}}{2} = \frac{4 \pm \sqrt{-36}}{2} = \frac{4 \pm 6i}{2} = 2 \pm 3i$$

The general solution is $y = e^{2x}(C_1 \cos 3x + C_2 \sin 3x)$

~~Ex~~ solve $(D^3 - D^2 - 8D + 12)y = 0$

Soln let $y = e^{mx}$

$$\therefore (D^3 - D^2 - 8D + 12)e^{mx} = 0$$

$$(m^3 - m^2 - 8m + 12)e^{mx} = 0$$

$$A.E \text{ is } m^3 - m^2 - 8m + 12 = 0$$

$$\Rightarrow m^3 - 2m^2 + m^2 - 2m - 6m + 12 = 0$$

$$\Rightarrow m^2(m-2) + m(m-2) - 6(m-2) = 0$$

$$\Rightarrow (m-2)(m^2+m-6) = 0$$

$$\Rightarrow (m-2)(m^2+3m-2m-6) = 0$$

$$\Rightarrow (m-2)\{m(m+3)-2(m+3)\} = 0$$

$$\Rightarrow (m-2)(m+3)(m-2) = 0$$

$$\therefore m = 2, 2, -3$$

General solution is $y = (c_1 + c_2 x)e^{2x} + c_3 e^{-3x}$

Q. Solve $(D^3 - 5D^2 + 7D - 3)y = 0$

Soln: let $y = e^{mx} = 0$

$$\therefore (D^3 - 5D^2 + 7D - 3)e^{mx} = 0$$

$$(m^3 - 5m^2 + 7m - 3)e^{mx} = 0$$

$$\therefore \text{A.E. } m^3 - 5m^2 + 7m - 3 = 0$$

$$\Rightarrow m^3 - m^2 - 4m^2 + 4m + m - 3 = 0$$

$$\Rightarrow m^2(m-1) - 4m(m-1) + 3(m-1) = 0$$

$$\Rightarrow (m-1)(m^2 - 4m + 3) = 0$$

$$\Rightarrow (m-1)(m^2 - 3m - m + 3) = 0$$

$$\Rightarrow (m-1)\{m(m-3) - 1(m-3)\} = 0$$

$$\Rightarrow (m-1)(m-3)(m-1) = 0$$

$$\therefore m = 1, 1, 3$$

The general solution is $y = (c_1 + c_2 x)e^x + c_3 e^{3x}$

Non-homogeneous Linear differential equations with constant coefficients

Standard form $(D^n + a_1 D^{n-1} + \dots + a_n)y = X$

D = Differentiation, $\frac{1}{D}$ = Integration

Working rule for finding the particular integral $f(D)y = X$

Some series:

$$(i) (1-D)^{-1} = 1 + D + D^2 + \dots + D^{n-1} + \dots$$

$$(ii) (1+D)^{-1} = 1 - D + D^2 - \dots + (-1)^n D^{n-1} + \dots$$

$$(iii) (1-D)^{-2} = 1 + 2D + 3D^2 + \dots + n D^{n-1} + \dots$$

$$(iv) (1+D)^{-2} = 1 - 2D + 3D^2 - \dots + (-1)^n n D^{n-1} + \dots$$

Type 1 when $X = x^m$

Question: $(D^2 - 4)y = x^2$

Soln Here A.E. $m^2 - 4 = 0$
 $m = \pm 2$

$$C.F. y_c = c_1 e^{-2x} + c_2 e^{2x}$$

$$\begin{aligned} P.I: y_p &= \frac{1}{D^2 - 4} x^2 \\ &= \frac{1}{-4(1 - \frac{D^2}{4})} x^2 \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{-4} \left(1 - \frac{D^2}{4} \right)^{-1} x^2 \\
&= -\frac{1}{4} \left(1 + \frac{D^2}{4} + \dots \right) x^2 \\
&= -\frac{1}{4} \left(x^2 + \frac{2}{4} \cdot 2x \right) \\
&= -\frac{1}{4} x^2 - \frac{1}{8} x
\end{aligned}$$

The general solution is $y = y_c + y_p$

$$= c_1 e^{2x} + c_2 e^{-2x} - \frac{1}{4} x^2 - \frac{1}{8} x$$

Question 2 Solve $(D^2 - 4D + 4)y = x^3$

Solution: Here, A.E. $m^2 - 4m + 4 = 0$

$$\Rightarrow (m-2)^2 = 0$$

$$\therefore m = 2, 2$$

$$C.F. y_c = (c_1 + c_2 x) e^{2x}$$

$$\text{Now, P.I. } y_p = \frac{1}{D^2 - 4D + 4} x^3$$

$$= \frac{1}{(2-D)^2} x^3$$

$$= \frac{1}{4(1 - \frac{D}{2})^2} x^3$$

$$= \frac{1}{4} \left(1 - \frac{D}{2} \right)^{-2} x^3$$

$$= \frac{1}{4} \left(1 + 2 \frac{D}{2} + 3 \frac{D^2}{4} + 4 \frac{D^3}{8} + \dots \right) x^3$$

$$= \frac{1}{4} \left(1 + D + \frac{3}{4} D^2 + \frac{D^3}{2} + \dots \right) x^3$$

$$= \frac{1}{4} \left(x^3 + 3x^2 + \frac{3}{4} \cdot 6x + \frac{1}{2} \cdot 6 \right)$$

$$= \frac{1}{4} \left(x^3 + 3x^2 + \frac{9}{2} x + 3 \right)$$

The general solⁿ is $y = y_c + y_p$

$$= (c_1 + c_2 x) e^{2x} + \frac{1}{4} (x^3 + 3x^2 + 2x + 3) \quad \text{A}$$

the general soln is $y = y_c + y_p$
 $= (c_1 + c_2 x) e^{2x} + \frac{1}{4} (x^3 + 3x^2 + \frac{9}{2}x + 3)$

Solve $(2D^2 + 5D + 2)y = x^2$

Soln Here, $(2D^2 + 5D + 2)y = x^2$

Let $y = e^{mx}$

$\therefore (2D^2 + 5D + 2) e^{mx} = x^2$

A.E $2m^2 + 5m + 2 = 0$

$\Rightarrow 2m^2 + 4m + m + 2 = 0$

$\Rightarrow 2m(m+2) + 1(m+2) = 0$

$\Rightarrow (m+2)(2m+1) = 0$

$\therefore m = -2, -\frac{1}{2}$

C.F $y_c = c_1 e^{-2x} + c_2 e^{-\frac{1}{2}x}$

Now P.I $y_p = \frac{1}{(2D^2 + 5D + 2)} x^2$

$= \frac{1}{2(1 + (\frac{5}{2}D + D^2))} x^2$

$= \frac{1}{2} [1 + (\frac{5}{2}D + D^2)]^{-1} x^2$

$= \frac{1}{2} (1 - \frac{5}{2}D - D^2 + \frac{25}{4}D^2 + \dots) x^2$

$= \frac{1}{2} (x^2 - \frac{5}{2}D x + \frac{21}{4}D^2 x + \dots)$

$= \frac{1}{2} (x^2 - 5x + \frac{21}{2})$

the general soln
 $y = y_c + y_p$
 $y = c_1 e^{-2x} + c_2 e^{-\frac{1}{2}x} + \frac{1}{2} (x^2 - 5x + \frac{21}{2})$

Type II: When $\chi = e^{ax}$

(i) if $y_p = \frac{1}{f(D)} e^{ax}$ then put $D = a$ when $f(a) \neq 0$

(ii) if $f(a) = 0$ then

$$\frac{1}{f(D)} e^{ax} = x \frac{1}{f'(D)} e^{ax} = x \frac{1}{f'(a)} e^{ax}; f'(a) \neq 0$$

(iii) If $f'(a) = 0$ then

$$\begin{aligned} \frac{1}{f(D)} e^{ax} &= x \frac{1}{f'(D)} e^{ax} = x^2 \frac{1}{f''(D)} e^{ax} \\ &= x^2 \frac{1}{f''(a)} e^{ax} \text{ when } f''(a) \neq 0 \end{aligned}$$

Question 1 Solve $(D^2 - 3D + 2)y = e^{5x}$

Soln: Here, $(D^2 - 3D + 2)y = e^{5x}$

$$\text{Let } y = e^{mx}$$

$$\therefore (D^2 - 3D + 2)e^{mx} = e^{5x}$$

$$\text{A.E. } m^2 - 3m + 2 = 0$$

$$\Rightarrow m^2 - 2m - m + 2 = 0$$

$$\Rightarrow m(m-2) - 1(m-2) = 0$$

$$\Rightarrow (m-2)(m-1) = 0$$

$$\therefore m = 1, 2$$

$$\text{C.F. } y_c = c_1 e^x + c_2 e^{2x}$$

Type II: When $x = e^{ax}$

(i) if $y_p = \frac{1}{f(D)} e^{ax}$ then put $D = a$ when $f(a) \neq 0$

(ii) if $f(a) = 0$ then

$$\frac{1}{f(D)} e^{ax} = x \frac{1}{f'(D)} e^{ax} = x \frac{1}{f'(a)} e^{ax}; f'(a) \neq 0$$

(iii) If $f'(a) = 0$ then

$$\begin{aligned} \frac{1}{f(D)} e^{ax} &= x \frac{1}{f'(D)} e^{ax} = x^2 \frac{1}{f''(D)} e^{ax} \\ &= x^2 \frac{1}{f''(a)} e^{ax} \text{ when } f''(a) \neq 0 \end{aligned}$$

Question 1 Solve $(D^2 - 3D + 2)y = e^{5x}$

Soln: Here, $(D^2 - 3D + 2)y = e^{5x}$

$$\text{Let } y = e^{mx}$$

$$\therefore (D^2 - 3D + 2)e^{mx} = e^{5x}$$

$$\text{A.E. } m^2 - 3m + 2 = 0$$

$$\Rightarrow m^2 - 2m - m + 2 = 0$$

$$\Rightarrow m(m-2) - 1(m-2) = 0$$

$$\Rightarrow (m-2)(m-1) = 0$$

$$\therefore m = 1, 2$$

$$\text{C.F. } y_c = c_1 e^x + c_2 e^{2x}$$

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$$\text{Now P.I. } y_p = \frac{1}{D^2 - 3D + 2} e^{5x}$$

$$= \frac{1}{5^2 - 3 \cdot 5 + 2} e^{5x}$$

$$= \frac{1}{12} e^{5x}$$

The general soln $y = y_h + y_p$

$$y = C_1 e^x + C_2 e^{2x} + \frac{1}{12} e^{5x} \quad A$$

Question 2 Solve $(D^2 - 3D + 2)y = e^x$

Soln: Here $(D^2 - 3D + 2)y = e^x$

$$A.E \quad m^2 - 3m + 2 = 0$$

$$\Rightarrow m^2 - 2m - m + 2 = 0$$

$$\Rightarrow m(m-2) - 1(m-2) = 0$$

$$\Rightarrow (m-2)(m-1) = 0$$

$$\therefore m = 1, 2$$

$$\text{C.F., } y_c = c_1 e^x + c_2 e^{2x}$$

$$\text{Now P.I., } y_p = \frac{1}{D^2 - 3D + 2} e^x$$

$$= x \frac{1}{2D - 3} e^x$$

$$= x \frac{1}{2(1) - 3} e^x$$

$$= -x e^x$$

The general solution is $y = y_c + y_p$

$$y = c_1 e^x + c_2 e^{2x} - x e^x$$

Case III: If $x = \sin ax / \cos ax$

(i) $y_p = \frac{1}{f(D)} \sin ax$ then put $D^2 = -a^2$ when $f(-a^2) \neq 0$

(iii) If $f(-a^2) = 0$ then,

$$\frac{1}{f(D)} \sin ax / \cos ax = x \frac{1}{f'(D)} \sin ax / \cos ax$$

OR, $\frac{1}{D^2 + a^2} \cos ax = \frac{x}{2a} \sin ax$

$\frac{1}{D^2 + a^2} \sin ax = \frac{x}{2a} \cos ax$

Question 1: Solve $(D^2 - 3D + 2)y = \sin 3x$

Solⁿ: Here $(D^2 - 3D + 2)y = \sin 3x$

A.E $m^2 - 3m + 2 = 0$

$$\Rightarrow m^2 - 2m - m + 2 = 0$$

$$\Rightarrow m(m-2) - 1(m-2) = 0$$

$$\Rightarrow (m-2)(m-1) = 0$$

$$\therefore m = 1, 2$$

C.F, $y_c = c_1 e^x + c_2 e^{2x}$

P.I, $y_p = \frac{1}{(D^2 - 3D + 2)} \sin 3x$

$$= \frac{1}{-3^2 - 3D + 2} \sin 3x$$

$$= \frac{1}{-9D - 2} \sin 3x$$

$$= -\frac{1}{9D + 2} \sin 3x$$

$$= -\frac{(3D - 2)}{(3D + 2)(3D - 2)} \sin 3x$$

$$= -\frac{(3D - 2)}{9D^2 - 2^2} \sin 3x$$

$$= -\frac{3D - 2}{9(-3^2) - 4} \sin 3x$$

$$= -\frac{3D - 2}{-130} \sin 3x$$

$$= \frac{1}{130} (3 \cos 3x - 2 \sin 3x)$$

$\therefore$ The general solution is $y = y_c + y_p$
 $= c_1 e^x + c_2 e^{2x} + \frac{1}{130} (9 \cos 2x - 7 \sin 2x)$

Question 2 Solve $(D^2 + 4)y = \cos 2x$ *

Soln: Here $(D^2 + 4)y = \cos 2x$

A.E $m^2 + 4 = 0$

$m^2 = -4$

$\therefore m = \pm 2i$

C.F, $y_c = c_1 \cos 2x + c_2 \sin 2x$

Now P.I $y_p = \frac{1}{D^2 + 4} \cos 2x$

$= x \frac{1}{2D + 4} \cos 2x$

$= \frac{x}{2} \sin 2x \cdot \frac{1}{2}$

$= \frac{x}{4} \sin 2x$

$\therefore$ The general soln is $y = y_c + y_p$

$= c_1 \cos 2x + c_2 \sin 2x + \frac{x}{4} \sin 2x$